

SKILL – 7

2520090160 – Snigdha

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <unistd.h>
```

```
#include <sys/wait.h>
```

```
int main()
```

```
{
```

```
    pid_t pid;
```

```
    int status;
```

```
    printf("Parent PID: %d\n", getpid());
```

```
    pid = fork();
```

```
    if (pid < 0)
```

```
    {
```

```
        perror("fork failed");
```

```
        return 1;
```

```
    }
```

```
    if (pid == 0)
```

```
    {
```

```
        printf("Child PID: %d\n", getpid());
```

```
        printf("Child: Doing some work...\n");
```

```

    sleep(3);

    printf("Child: Work completed.\n");
    exit(42);
}
else
{
    printf("Parent: Waiting for child using waitpid()...\n");

    waitpid(pid, &status, 0);

    if (WIFEXITED(status))
    {
        printf("Parent: Child exited normally.\n");
        printf("Parent: Child exit status = %d\n",
            WEXITSTATUS(status));
    }
}

return 0;
}

```

```

snigdha@Snigdha:~/quotes_practical$ nano waitpid.c
snigdha@Snigdha:~/quotes_practical$ gcc -Wall -Wextra -o waitpid waitpid.c
snigdha@Snigdha:~/quotes_practical$ ./waitpid
Parent PID: 1038
Parent: Waiting for child using waitpid()...
Child PID: 1039
Child: Doing some work...
Child: Work completed.
Parent: Child exited normally.
Parent: Child exit status = 42
snigdha@Snigdha:~/quotes_practical$ |

```

```
#include <stdio.h>

#include <stdlib.h>

#include <string.h>

#include <unistd.h>

#include <sys/stat.h>
```

```
int main(int argc, char *argv[])
```

```
{
```

```
    char *path;
```

```
    char *path_copy;
```

```
    char *dir;
```

```
    char full_path[1024];
```

```
    struct stat file_info;
```

```
    if (argc != 2)
```

```
    {
```

```
        printf("Usage: %s <command>\n", argv[0]);
```

```
        return 1;
```

```
    }
```

```
    path = getenv("PATH");
```

```
    if (path == NULL)
```

```
    {
```

```
        printf("PATH variable not found.\n");
```

```
        return 1;
```

```
    }
```

```
path_copy = strdup(path);

if (path_copy == NULL)
{
    perror("strdup");
    return 1;
}

printf("Searching for command: %s\n\n", argv[1]);

dir = strtok(path_copy, ":");

while (dir != NULL)
{
    snprintf(full_path, sizeof(full_path),
             "%s/%s", dir, argv[1]);

    if (stat(full_path, &file_info) == 0)
    {
        if (access(full_path, X_OK) == 0)
        {
            printf("Executable found: %s\n", full_path);

            free(path_copy);
            return 0;
        }
    }
}
```

```

    dir = strtok(NULL, ":");
}

printf("Command '%s' not found in PATH.\n", argv[1]);

free(path_copy);

return 1;
}

```

```

snigdha@Snigdha:~/quotes_practical$ nano path_search.c
snigdha@Snigdha:~/quotes_practical$ gcc -Wall -Wextra -o path_search path_search.c
snigdha@Snigdha:~/quotes_practical$ ./path_search ls
Searching for command: ls

Executable found: /usr/bin/ls
snigdha@Snigdha:~/quotes_practical$ ./path_search gcc
Searching for command: gcc

Executable found: /usr/bin/gcc
snigdha@Snigdha:~/quotes_practical$ ./path_search abcxyz
Searching for command: abcxyz

Command 'abcxyz' not found in PATH.
snigdha@Snigdha:~/quotes_practical$ echo $PATH
/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/u
sr/lib/wsl/lib:/mnt/c/Program Files/WindowsApps/MicrosoftCorporationII.WindowsSubsystemForL
inux_2.7.11.0_x64__8wekyb3d8bbwe:/mnt/c/Program Files/Python314/Scripts:/mnt/c/Program Fil
es/Python314:/mnt/c/Program Files/Common Files/Oracle/Java/javapath:/mnt/c/Program Files (
x86)/Common Files/Oracle/Java/java8path:/mnt/c/Program Files (x86)/Common Files/Oracle/Java
/javapath:/mnt/c/Windows/system32:/mnt/c/Windows:/mnt/c/Windows/System32/Wbem:/mnt/c/Window
s/System32/WindowsPowerShell/v1.0:/mnt/c/Windows/System32/OpenSSH:/mnt/c/Program Files/Gi
t/cmd:/mnt/c/Program Files/dotnet:/mnt/c/Program Files/nodejs:/mnt/c/Program Files/MySQL/
MySQL Shell 8.0/bin:/mnt/c/Users/snigd/AppData/Local/Microsoft/WindowsApps:/mnt/c/Users/sn
igd/AppData/Local/Programs/Microsoft VS Code/bin:/mnt/c/Users/snigd/AppData/Roaming/npm:/mn
t/c/Users/snigd/AppData/Local/Python/bin:/mnt/c/Users/snigd/AppData/Local/Programs/Antigrav
ity/bin:/snap/bin
snigdha@Snigdha:~/quotes_practical$ ls -l $(which ls)
lrwxrwxrwx 1 root root 29 Mar 30 2026 /usr/bin/ls -> ../lib/cargo/bin/coreutils/ls
snigdha@Snigdha:~/quotes_practical$

```